

TOOLARA, Australia

WMO: 945580

Lat: 25.971S

Lon: 152.861E

Elev: 469

StdP: 14.45

Time Zone: 10.00 (E10)

Period: 01-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	47.9	50.0	28.3	22.6	60.3	31.7	26.5	59.9	17.6	61.0	15.6	57.7	9.9	240	0.349

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.2	91.7	73.8	89.0	73.0	86.8	72.1	77.3	86.0	76.1	83.8	75.1	82.0	10.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	134.4	80.3	74.1	129.5	79.2	73.1	125.0	78.2	41.0	85.9	39.9	83.5	38.8	82.0	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.4	16.7	14.2	DB	44.1	98.2	2.2	2.7	42.5	100.1	41.2	101.6	39.9	103.1	38.3	105.0
			WB	38.4	80.3	1.9	1.2	37.0	81.1	35.9	81.8	34.9	82.5	33.5	83.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.0	77.3	77.1	74.3	71.0	65.3	61.7	60.8	63.1	67.8	71.4	74.0	76.5	(6)
	DBStd	6.90	3.60	3.32	3.06	3.03	3.30	3.90	3.19	3.72	4.20	4.23	3.56	4.30	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	387	0	0	0	1	36	111	135	82	19	3	0	0	(9)
	CDD50	7290	847	758	754	629	474	351	334	406	535	662	719	821	(10)
	CDD65	2201	382	338	289	180	45	12	4	23	103	200	269	356	(11)
	CDH74	12346	2549	2174	1360	602	89	11	5	83	495	1068	1521	2389	(12)
	CDH80	3702	875	708	325	76	3	0	0	8	100	293	424	890	(13)
(14) Wind	WSAvg	7.7	8.5	8.0	8.4	7.7	7.6	7.4	7.1	7.2	7.2	7.9	7.6	7.7	(14)
(15) Precipitation	PrecAvg	50.7	7.3	8.1	6.5	4.0	3.9	3.5	2.1	2.8	1.9	2.9	3.1	5.9	(15)
	PrecMax	75.2	17.6	20.0	18.9	9.8	12.0	8.7	6.7	17.4	7.0	7.0	6.1	19.3	(16)
	PrecMin	34.0	0.7	1.1	1.7	1.3	1.0	0.7	0.5	0.0	0.0	0.7	0.3	2.2	(17)
	PrecStd	11.1	4.6	5.7	4.6	2.0	2.5	2.3	1.5	4.1	1.9	1.8	1.5	4.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.8	94.5	89.9	85.4	79.3	75.7	74.5	80.1	87.1	90.8	91.1	96.0	(19)
		MCWB	76.3	76.0	74.5	70.5	64.9	65.8	59.9	63.9	64.7	68.0	71.3	74.7	(20)
	2%	DB	90.6	89.4	86.2	82.0	76.4	72.7	71.7	76.2	82.8	86.2	87.4	91.4	(21)
		MCWB	74.6	75.2	72.4	69.0	63.2	63.2	59.6	61.4	65.3	67.4	70.8	73.8	(22)
	5%	DB	87.6	86.7	83.5	79.7	74.1	70.3	69.7	73.3	79.5	83.2	84.7	88.1	(23)
		MCWB	73.8	74.0	71.1	68.3	62.9	62.0	59.0	60.0	63.9	67.1	69.8	72.6	(24)
	10%	DB	85.1	84.4	81.0	77.6	71.8	68.2	67.5	70.7	76.5	80.0	82.0	85.0	(25)
		MCWB	72.7	73.1	70.5	67.9	62.5	61.1	58.9	58.7	62.7	66.1	68.6	71.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.7	79.5	77.3	74.6	68.4	68.4	67.5	67.7	70.1	73.2	75.1	78.3	(27)
		MCDB	89.2	89.8	84.7	79.7	73.7	71.7	70.5	74.7	79.1	84.4	85.7	89.6	(28)
	2%	WB	76.9	77.3	75.4	72.5	67.1	66.4	65.0	65.0	68.1	71.2	73.5	76.5	(29)
		MCDB	85.3	85.7	81.2	77.0	72.3	69.6	67.4	71.0	76.9	80.3	82.2	85.2	(30)
	5%	WB	75.6	76.0	73.9	70.9	65.7	64.9	62.5	63.3	66.6	69.8	72.0	75.0	(31)
		MCDB	82.9	83.4	79.1	76.1	70.4	67.9	66.1	68.8	74.2	77.5	80.1	82.7	(32)
	10%	WB	74.6	74.7	72.5	69.4	64.5	63.6	60.6	61.6	65.3	68.4	70.6	73.7	(33)
		MCDB	81.3	81.2	77.7	75.0	68.9	66.5	65.4	67.1	72.2	75.9	78.2	80.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	14.9	13.9	14.5	14.3	13.4	15.4	17.2	18.2	18.3	17.2	17.0	(35)
		MCDBR	19.3	18.5	17.8	17.2	16.8	14.7	16.9	19.5	21.9	22.5	20.9	21.9	(36)
	5% WB	MCWBR	6.4	6.6	6.7	7.1	7.9	8.2	8.7	8.0	8.2	8.1	7.2	7.7	(37)
		MCWBR	16.0	16.2	14.1	14.3	13.2	11.5	12.7	16.4	17.9	19.4	18.3	19.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.384	0.377	0.359	0.336	0.318	0.312	0.301	0.312	0.351	0.370	0.373	0.387	(40)
		taud	2.519	2.545	2.592	2.642	2.640	2.652	2.667	2.617	2.497	2.476	2.504	2.493	(41)
	Ebn at Noon		304	301	298	291	284	279	287	295	296	300	305	304	(42)
		Edh at Noon	36	34	31	27	25	23	24	27	34	36	36	37	(43)
(44) All-Sky Solar Radiation	RadAvg		2081	1898	1648	1440	1203	1021	1172	1452	1778	1957	2125	2120	(44)
	RadStd		169	164	123	99	63	77	101	113	136	165	193	232	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.67	+1.01	N/A	N/A	N/A	N/A	N/A	-126	+329	+198

Nomenclature: See separate page